

7.3

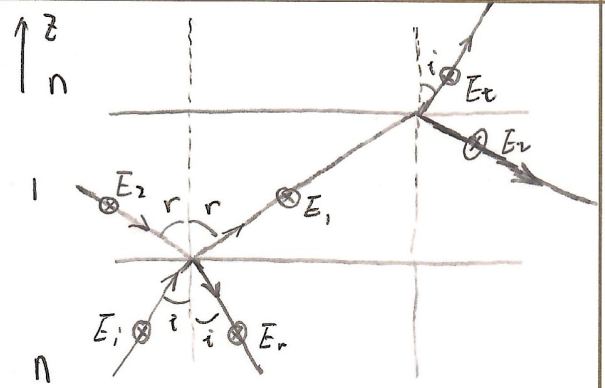
(a) Perpendicular to the incidence plane

The waves in the three regions are given by

$$\text{I: } \vec{E}(\vec{r}, t) = \vec{E} (E_i e^{i\vec{k} \cdot \vec{r} - i\omega t} + E_r e^{-i\vec{k} \cdot \vec{r} - i\omega t})$$

$$\text{II: } \vec{E}(\vec{r}, t) = \vec{E} (E_t e^{i\vec{k}' \cdot \vec{r} - i\omega t} + E_r e^{-i\vec{k}' \cdot \vec{r} - i\omega t})$$

$$\text{III: } \vec{E}(\vec{r}, t) = \vec{E} E_t e^{i\vec{k}' \cdot \vec{r} - i\omega t}$$



where $\vec{E}$ is the unit vector of the linear polarization, and $k = n\omega/c$, $k' = \omega/c$.

The boundary conditions at $z=0$ and $z=D$ lead to

$$E_i + E_r = E_t + E_r$$

$$n(E_i - E_r) \cos i = (E_t - E_r) \cos r$$

$$E_i e^{ikD \cos r} + E_r e^{-ikD \cos r} = E_t e^{ikD \cos i}$$

$$(E_i e^{ikD \cos r} - E_r e^{-ikD \cos r}) \cos r = n E_t e^{ikD \cos i} \cos i$$

From Snell's law, we know $\frac{\sin i}{\sin r} = \frac{1}{n}$, or $\sin r = n \sin i$, and

$$\cos r = (1 - n^2 \sin^2 i)^{1/2} \quad \text{Let } \alpha = n \cos i / \cos r = n \cos i / (1 - n^2 \sin^2 i)^{1/2}$$

$$\beta = k'D \cos r = \frac{\omega D/c}{(1 - n^2 \sin^2 i)^{1/2}} \quad \gamma = kD \cos i = n\omega D \cos i/c, \quad \text{then we have the equivalent equations}$$

$$E_i + E_r = E_t + E_r$$

$$\alpha(E_i - E_r) = E_t - E_r$$

$$E_i e^{i\beta} + E_r e^{-i\beta} = E_t e^{i\gamma}$$

$$E_i e^{i\beta} - E_r e^{-i\beta} = \alpha E_t e^{i\gamma}$$

$$E_t = \frac{1}{2}(1 + \alpha) E_i + \frac{1}{2}(1 - \alpha) E_r$$

$$E_r = \frac{1}{2}(1 - \alpha) E_i + \frac{1}{2}(1 + \alpha) E_t$$

$$E_t = \frac{1}{2}(1 + \alpha) E_i e^{i\gamma - i\beta}$$

$$E_r = \frac{1}{2}(1 - \alpha) E_i e^{i\gamma + i\beta}$$

$$\text{or } E_t = \frac{2\alpha e^{-i\gamma}}{2\alpha \cos \beta - i(1 + \alpha^2) \sin \beta} E_i$$

$$E_r = \frac{1 - \alpha^2}{2\alpha} \cdot \frac{i \sin \beta e^{i\gamma}}{2\alpha \cos \beta - i(1 + \alpha^2) \sin \beta} E_i = \frac{i(1 - \alpha^2) \sin \beta}{2\alpha \cos \beta - i(1 + \alpha^2) \sin \beta} E_i$$

$$\text{And, } R = \frac{(1 - \alpha^2)^2 \sin^2 \beta}{4\alpha^2 \cos^2 \beta + (1 + \alpha^2)^2 \sin^2 \beta}, \quad T = \frac{4\alpha^2}{4\alpha^2 \cos^2 \beta + (1 + \alpha^2)^2 \sin^2 \beta} \quad \text{Clearly, } R + T = 1.$$

For $i > \arcsin(\frac{1}{n})$, there will be total internal reflection, and α and β will become pure imaginary.

$$\alpha = \frac{i n \cos i}{(n^2 \sin^2 i - 1)^{1/2}} = i a, \quad \beta = \frac{i \omega D/c}{(n^2 \sin^2 i - 1)^{1/2}} = i b$$

Using the identities $\cos(ia) = \cosh a$, $\sin(ia) = i \sinh a$, the reflection and transmission coefficients will be

$$R = \frac{(1 + a^2)^2 \sinh^2 b}{4a^2 \cosh^2 b + (1 - a^2)^2 \sinh^2 b}, \quad T = \frac{4a^2}{4a^2 \cosh^2 b + (1 - a^2)^2 \sinh^2 b}$$

(b) Parallel to the incidence plane

Similarly to part (a), we can write down the corresponding equation according to the boundary conditions,

$$(E_i - E_r) \cos r = (E_t - E_r) \cos i$$

$$n(E_i + E_r) = E_t + E_r$$

$$(E_i e^{ik' \cos r} - E_r e^{-ik' \cos r}) \cos r = E_t e^{ik \cos i} \cos i$$

$$E_i e^{ik' \cos r} + E_r e^{-ik' \cos r} = n E_t e^{ik \cos i}$$

Let $\delta = \cos i / \cos r = \cos i / (1 - n^2 \sin^2 i)^{1/2}$, then we have

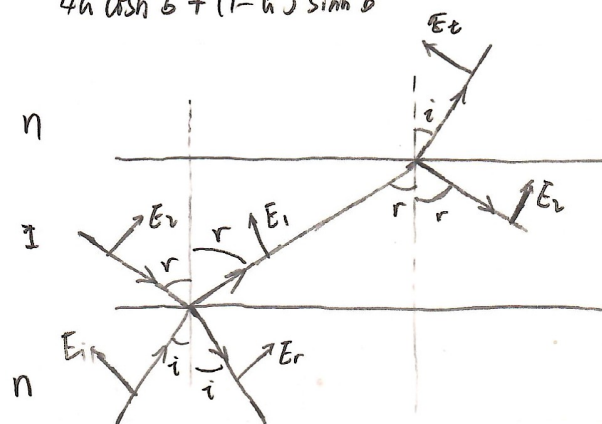
$$E_i = \frac{1}{2}(n + \delta) E_t + \frac{1}{2}(n - \delta) E_r = \frac{1}{2}(n + \delta) E_t e^{i(\gamma - \beta)}$$

$$E_r = \frac{1}{2}(n - \delta) E_t + \frac{1}{2}(n + \delta) E_r = \frac{1}{2}(n - \delta) E_t e^{i(\gamma + \beta)}$$

Which can be solved and gives

$$E_t = \frac{2n\delta e^{-i\gamma}}{2n\delta \cos \beta - i(n^2 + \delta^2) \sin \beta} E_i$$

$$E_r = \frac{i(n^2 - \delta^2) \sin \beta}{2n\delta \cos \beta - i(n^2 + \delta^2) \sin \beta} E_i$$



III

II

I

Therefore, $R = \left| \frac{\bar{E}_r}{\bar{E}_i} \right|^2 = \frac{(n^2 - d^2)^2 \sin^2 \beta}{4n^2 d^2 \cos^2 \beta + (n^2 + d^2)^2 \sin^2 \beta}$

$$T = \left| \frac{\bar{E}_t}{\bar{E}_i} \right|^2 = \frac{4n^2 d^2}{4n^2 d^2 \cos^2 \beta + (n^2 + d^2)^2 \sin^2 \beta}$$

Again, $R + T = 1$. For $i > \arcsin(1/n)$,

$$\delta = \frac{i \cos i}{(n^2 \sin^2 i - 1)^{1/2}} = id, \quad \beta = ib,$$

and

$$R = \frac{(n^2 + d^2)^2 \sinh^2 b}{4n^2 d^2 \cosh^2 b + (n^2 - d^2)^2 \sinh^2 b}$$

$$T = \frac{4n^2 d^2}{4n^2 d^2 \cosh^2 b + (n^2 - d^2)^2 \sinh^2 b}$$